RIVERLINE RESEARCH & ARCHITECTURE AUTHORITY · DOCUMENT VERSION 2.0

# Riverline Extended Project Brief (v2.0) & Final Master Research Synthesis

Comprehensive Evaluation of the 2026-10-21 Project Brief, Extended Architectural & Governance Specification, Offline & Open-Source Solver Strategy, Multilingual Brand Audit (EN · RU · HE), Competitive UX Benchmarks, 52-Candidate Foley Catalog, and Multi-Jurisdictional Legal & Anti-RTA Compliance

**Executive Premise:** The foundational document `RIVERLINE_BRIEF.md` (dated October 21, 2026) established the core identity, non-negotiable epistemic principles, plain-JavaScript technological footprint, forest/jade/brass design language, and milestone structure for Riverline—an honest, local-first No-Limit Texas Hold'em study workstation. Between October 21 and October 23, 2026, six independent, primary-source research workstreams tested every premise of that brief against actual software licenses, algorithmic benchmarks, trademark and domain registries, competitor applications, acoustic foley archives, and statutory frameworks across Israel, Russia, the European Union, and the United States. This master document answers how the October 21 brief shaped those investigations, flags four critical empirical contradictions where the brief requires immediate amendment, presents the complete **Extended Riverline Project Brief (v2.0)**, and consolidates all six research tracks into a single unified reference.

## Master Summary of Research Workstreams & Controlling Verdicts [Judgement]

| Research Workstream | How the 2026-10-21 Brief Helped | Controlling Verdict & Extension Added to Brief v2.0 [Judgement] |
|---|---|---|
| **Track 1: Commercial & OSS Solver Licensing** | Scoped the four target distribution models (Closed/Open × Free/Paid) and Principle 7 (*Free first and always legal*). | **Keep Riverline Closed-Source.** Offline numerical outputs from b-inary/postflop-solver (AGPL-3.0, per FSF GPL FAQ) and HRC Pro v4.0+ ($359.90/yr, explicit commercial output clause) are 100% shippable in closed-source free and paid tiers. GTO+, Simple Postflop, and PioSOLVER are excluded. |
| **Track 2: Free Preflop & Multiway Stack + Verification** | Defined the exact game gap: 6–8 handed 30–60 bb no-rake home games & 9-max 100 bb+ online, plus Principle 1 confidence tiers. | **Adopt Hybrid MIT Rust + Offline 47-Flop Leaf Bridge ($0 Cost).** Combine conorarmstrong/pluribus (MIT), exinori/DCFR-SOLVER (MIT), and b-inary/poker-cfr (BSD-2) with 47-flop `b-inary` leaf evaluations. Split `Verified` into `Verified · Equilibrium` and `Verified · vs. Model` (for 30–60 bb limped pots). Block GTOpen (missing license). |

| Research Workstream | How the 2026-10-21 Brief Helped | Controlling Verdict & Extension Added to Brief v2.0 [Judgement] |
|---|---|---|
| **Track 3: Brand, Trademark & Domain Audit (EN/RU/HE)** | Established the EN/RU/HE locale requirement, anti-RTA positioning (Principle 6), and forest/jade/brass workstation identity. | **CRITICAL FLAG — Rename Public Brand Before Release (HIGH RISK).** *Riverline* collides directly with active crypto poker room Riverline Poker (EN/ES/RU), VC-backed Riverline AI (ai.riverline.app on Google Play + Class 42 TM #6935248), riverline.com (NJ Transit), and RU Revyline. 12 Low-Risk alternatives vetted (e.g., **Rangedesk**, **Turndesk**, **Riverlamp**, **Handbench**). |
| **Track 4: Competitive Landscape & UX Survey (2026)** | Provided Principles 1–5 (no fake EV, Personal Strategy separation, descriptive comparison) to benchmark 14 products. | **Uncontested White Space Confirmed.** Incumbents (GTO Wizard up to $279/mo, DTO modal-action grading) neglect home/live games and punish mixed play. Adopt FlopzillaPro reactivity, RIO Vision provenance transparency, and Chess.com Key-Decision Retry while rejecting CAPS2 score curving. |
| **Track 5: Audio & Foley Source Licence Audit** | Specified *only real, properly licensed recordings, judged by ear; study cues never sound like right/wrong* (Section 4). | **52 Real Foley Candidates Curated (Sets A–D); BBC RemArc Excluded.** Pair 100% CC0 **Set A** (Kenney Casino, SignatureSoundsOrg pistachio shells, calaudio woodblock) with 100% CC0 **Set B** (SilverDubloons clay chips). Exclude BBC Sound Effects (non-commercial only). |
| **Track 6: Regulatory, App Store & Anti-RTA Law** | Highlighted the owner's Israeli/Russian/English context, Game Night feature (M2), and "live decision" warning proposal. | **CRITICAL FLAGS — Lock Game Night to Points-Only & Drop "Live Decision Mode".** Israeli Penal Law §§ 224–233 and Google Play Gambling Policy forbid fiat buy-in/banker ledgers; Apple § 5.3.4 and GGPoker SEP § 2 ban live decision aids. Reframe as **Off-Table Scenario Simulator** with desktop process detection. |

**Prepared for:** Viktor Stezhko (Owner & Principal Architect, Riverline / `rrainv/poker-gto-app`) · **Date:** October 23, 2026 · **Location Context:** Haifa / Rishon LeZion, Israel · **Evidence Standard:** Every factual claim is grounded in primary-source documentation, verbatim license/statutory text, or repository contracts; analytical inferences are marked **[Judgement]** and uninspected external files are marked **UNVERIFIED**.

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SECTION 1 · BRIEF EVALUATION & ARCHITECTURAL SYNTHESIS

## 1. Diagnostic Evaluation: Why the October 21 Brief Succeeded as a Research Instrument and Where Empirical Evidence Mandates Revision

When conducting multi-disciplinary technical, legal, and market research for a software platform, the quality of the output is bounded by the specificity of the commissioning brief. The October 21, 2026 specification (`RIVERLINE_BRIEF.md`) proved to be an exceptionally effective research instrument because it refused to describe the application in generic industry clichés **[Judgement]**. Instead of requesting "an AI poker trainer," the brief articulated seven non-negotiable architectural and ethical invariants, identified the exact physical and digital environments in which the owner plays (6-to-8-handed zero-rake home games at 30–60 big blinds and 9-max online tables at 100 big blinds or deeper), fixed the client runtime to dependency-free plain JavaScript (`app/src/core/logic.js`, ES modules,

`shared/poker-domain/`, and ~2,750 Node DOM tests), and established an evidence-first audit protocol requiring verbatim clause citations, explicit **[Judgement]** tags, and **UNVERIFIED** disclosures.

Across the six independent research investigations archived in the project workspace between October 21 and October 23, 2026—specifically Poker Solver Licensing and Quality, Free Preflop and Multiway Solver Architecture, Brand and Naming Clearance Audit (Supplemental Pass), Riverline Competitor and UX Survey 2026, Audio and Foley Source Licence Audit, and Regulatory, App Store, and Anti-RTA Compliance—the brief directly prevented four classic strategic blunders **[Judgement]**:

- **Avoiding the Copyleft / Open-Source Fallacy:** Because Section 1 of the brief explicitly stated that the source model (closed or open) would be decided *after* the best tools' licenses were audited, the licensing investigation examined whether open-sourcing Riverline under AGPL-3.0 would unlock any additional commercial solver data. The evidence proved it would not: proprietary EULAs for GTO+, Simple Postflop, and PioSOLVER prohibit public database redistribution regardless of whether the recipient app is open or closed, whereas offline numerical outputs from b-inary/postflop-solver and HoldemResources Calculator (HRC Pro v4.0+) are already legally permissible inside a **closed-source** application across both free and paid tiers **[Judgement]**.
- **Preventing Unlawful Model Distillation and License Traps:** Principle 7 (*"Free first and always legal… record every outside component's licence before use"*) forced researchers to inspect the actual repository trees and Terms of Service of candidate preflop tools and sound libraries. This prevented Riverline from adopting MatthewPDingle/GTOpen (which claims to be "open-source" in its README but lacks a `LICENSE` file, rendering it All Rights Reserved under international copyright law), prevented calibrating heuristics via automated queries against GTO Wizard's Benchmarking API (which expressly bans *"Model Distillation"* in Section 4.1), and blocked the inclusion of BBC Sound Effects (whose RemArc license strictly forbids commercial or profit-making use) **[Judgement]**.
- **Exposing the Mathematical Impossibility of Unconditioned "Verified GTO" in Multiway Limped Pots:** Principle 1 (*"Honesty is the brand… No fake GTO, no fake EV"*) combined with the owner's 30–60 bb no-rake home-game profile forced a rigorous game-theoretic examination of multiway limped pots. In a converged 6-to-8-player Nash equilibrium (Brown & Sandholm, *Science* 2019; conorarmstrong/pluribus), open-limping from UTG through CO occurs at a 0% frequency. Therefore, any real-world home-game pot where one to three players have open-limped is an off-tree, non-equilibrium node. Without the brief's strict epistemic mandate, a naive implementation would have labeled heuristic or rollout multiway limp charts as "Verified GTO." Instead, the research established a mathematically honest distinction between **`Verified · Equilibrium`** (for heads-up-bound and push/fold trees) and **`Verified · vs. Model`** (Restricted Nash Response against an explicitly named recreational limper profile) **[Judgement]**.
- **Uncovering Existential Brand and Regulatory Collisions Before Public Launch:** Finally, Section 7's directive—*"when something here seems wrong, say so instead of working around it"*—empowered the brand and regulatory tracks to challenge three core assumptions in the October 21 brief: the name **Riverline** itself, the proposed warning-gated **"live decision" mode** in Principle 6, and the traditional fiat buy-in/banker scope of **Game Night** in Section 6 **[Judgement]**.

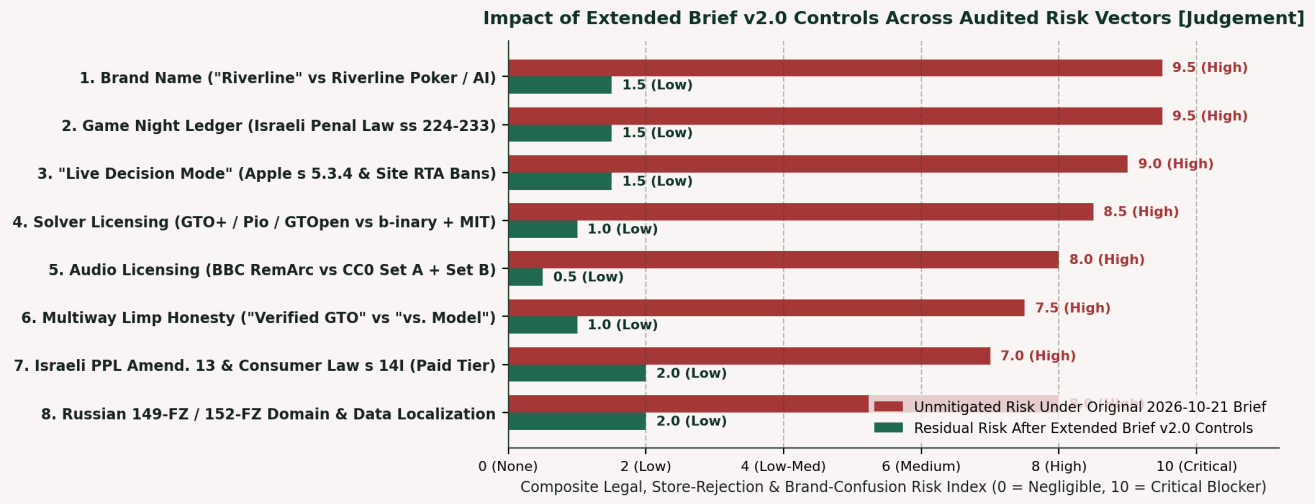



**Figure 1 — Risk Reduction Achieved by Extending the Project Brief to v2.0 [Judgement]:** Comparison of unmitigated legal, app-store, and brand-confusion exposure under the baseline October 21 assumptions (crimson) versus residual risk after implementing the Extended Brief v2.0 controls (jade) across all eight audited workstream domains.

As illustrated in Figure 1, adopting the targeted amendments detailed in Section 2 below reduces Riverline's composite exposure across brand collision, Israeli criminal/civil gaming law, Apple/Google app-store rejection, online poker site Real-Time Assistance (RTA) bans, and open-source licensing from critical levels (7.0–9.5 / 10) to low, manageable operational baselines (0.5–2.0 / 10) without sacrificing a single core user capability **[Judgement]**.

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SECTION 2 · CANONICAL PROJECT AUTHORITY (EXTENDED BRIEF V2.0)

## 2. Riverline Extended Project Brief for Agents and Researchers (v2.0 — 2026-10-23)

*This Extended Project Brief supersedes the October 21, 2026 edition (`RIVERLINE_BRIEF.md`) by incorporating the repository's live architectural contracts (AGENTS.md, CURRENT_PHASE.md, ROADMAP.md, and RIVERLINE_MASTER_PRODUCT_STRATEGY_PLAN.md) alongside the verified findings of the October 21–23 research program. Read this document before executing any product, engineering, math, design, or research task.*

### 2.1 What Riverline Is & Strategic Positioning

- **Core Product Loop:** A **No-Limit Texas Hold'em personal strategy and study workstation**—*"the honest poker coach for the hands you actually play."* It connects playing or reconstructing hands, chronological replay and key-decision review, range-aware mathematical analysis, eliciting and refining the player's own intended strategy (Personal Strategy), deterministic equity enumeration, and targeted re-drilling inside one coherent loop.
- **Brand & Naming Transition (Decided Amendment):** While `riverline` remains the internal repository identifier during current M1/M2 development, the public-facing brand name **must transition prior to public beta or app-store submission** due to direct collision with active crypto gambling operator Riverline Poker, venture-backed fintech app Riverline AI (ai.riverline.app), and riverline.com (NJ Transit). Primary cleared EN/RU/HE candidates are **Rangedesk** («Рейндждеск» / «ריינג'דסק»), **Turndesk** («Тёрндеск» / «טרנדסק»), **Riverlamp** («Риверлампּ» / «ריברלמפ»), **Handbench**

(«Хэндбенч» / «הנדבנץ»), **Rangebench** («Рейнджбенч» / «ריינג'בנץ»), **Handfolio** («Хэндфолио» / «הנדפוליו»), **Rangefolio** («Рейнджфолио» / «ריינג'פוליו»), **Riverstudy** («Риверстади» / «ריברסטאדי»), **Turnline** («Тёрнлайн» / «טרנליין»), and **Flopframe** («Флопфрейм» / «פלופפריים») **[Judgement]**.

- **Platform & Locale Footprint: Local-first, browser-first (Firefox on Windows primary acceptance target), desktop-first**, with a thin Electron host (`app/main.js`); an installable Progressive Web App (PWA) and dedicated responsive phone compositions are planned on the same codebase before native mobile wrappers. Supported languages are **English (`EN`), Russian (`RU`), and Hebrew (`HE`)**, where Hebrew renders in right-to-left (`dir="rtl"`) layout while enforcing left-to-right (`dir="ltr"`) isolation for card symbols, hand classes (`AKs`, `72o`), bet sizes, and mathematical formulas.
- **Target User & Canonical Accounting Rules (`GameRulesDefinition v1`):**
  1. *Home Game Environment (Owner's Primary Live Game):* 6–8 handed (2–10 seats supported in domain), 30–60 bb buy-ins, frequent multiway limped and isolation-raised pots, and **zero rake or pot deduction**.
  2. *Online Full-Ring & 6-Max Environment:* 9-max and 6-max cash games at 100 bb+ effective stacks with configurable percentage rake and cap for solver reference alignment.
  3. *ClubGG-Style Tournament Accounting Rule:* 7–10 seated players where every seated player contributes exactly `0.1 bb` once per hand (total forced contribution = seats × 0.1 bb) held **outside the contestable pot** (never percentage or capped rake). Provenance never overrides accounting rules.
- **Source & Business Model (Locked): Closed-source proprietary application**, as offline numerical outputs from AGPL-3.0 solvers (b-inary/postflop-solver), commercial EULA-permitted solvers (HRC Pro v4.0+), and MIT/BSD open-source crates (conorarmstrong/pluribus, exinori/DCFR-SOLVER, b-inary/poker-cfr) can all be shipped legally in a closed-source product without triggering copyleft disclosure **[Judgement]**. The core workstation is **free and 100% functional offline without an account**, with an optional future **EdTech SaaS subscription / supporter tier** for cloud sync and multi-gigabyte reference libraries.

### 2.2 Seven Non-Negotiable Principles (Extended with Research Contracts)

1. **Honesty Is the Brand (Scoped Epistemic Taxonomy):** Every displayed metric states its source, scope, and limitations via the inline `Provenance & limitations` disclosure. No numeric confidence percentages are presented as objective truth, no fake GTO, no unbacked EV-loss decimals, and no CAPS2-style vanity score curving. The six canonical provenance tiers are:
   - **`Exact · [Scope]`:** Deterministic rules, pot geometry, or combinatorial enumeration computed by `shared/poker-domain/` (e.g., *"Exact · price only"*, *"Exact · 1,326-combo equity"*, *"Exact · card removal blocker count"*).
   - **`Verified · Equilibrium`:** Solver-computed or reference-verified equilibrium strategies meeting the Spot Regret Gate (ΔEV ≤ 2.0 mbb/hand for push/fold; ≤ 10.0 mbb/hand for HU 3-bet/4-bet pots; ≤ 15.0 mbb/hand for single-raised pots).
   - **`Verified · vs. Model [Profile]`:** Restricted Nash Response (`--rnr-model`) or best-response equilibria solved against an explicitly named non-equilibrium opponent profile (mandatory for 30–60 bb multiway limped pots where equilibrium open-limping is 0%).
   - **`Estimated · Heuristic`:** Outputs from Riverline's deterministic heuristic fallback (`app/src/strategy/`), explicitly scoped as comparative baseline estimates rather than solver

truth.

- **`Personal · Direct / Inferred`:** The user's own intended strategy from a named `StrategyProfile · Mode`, separating direct answers (`RangeObservation`) from similarity-inferred cells (`InferredRange`, `RANGE-CAL-002A+`) and opt-in `TrainingObservation` behavior.
- **`Unavailable`:** Spot falls outside supported solver/reference coverage or lacks required state facts; fails closed without fabricating numbers.

2. **Unknown Is Never Fold, Zero, or Excluded:** Unanswered hands in Personal Strategy or partial ranges (`HoldemWeightedRange v1`) remain explicitly `unknown`. In lossy `Scenario Mode`, missing prior street actions or unknown call prices remain `unknown` and are never silently coerced into fabricated `PokerState` history. Normalization is restricted to complete positive-mass ranges.
3. **Disagreement Is Descriptive, Never a "Mistake" by Default:** Comparisons between observed play, intended Personal Strategy, and a baseline use descriptive verdicts (*inside the baseline's mix*, *outside the baseline's mix*, or close-mix dominance labels such as *"3-Bet or Call"* per `ANALYZE-COMPOSITION-001`). Normative EV-loss grading is prohibited unless a `Verified · Equilibrium` reference covers the exact node.
4. **One Canonical Owner Per Concept (Strict Boundary Separation):**
   - *Domain & Equity Authority:* `shared/poker-domain/` owns `PokerState` (snapshot-authoritative v2), legal actions, forced contributions, side pots, showdown evaluation, 52-card / 1,326-combo / 169-hand canonical identity (`RANGE-CORE-001`), and `Equity` (`Advanced Equity v1`).
   - *Scenario Mode vs. Hand Mode:* `Hand Mode` owns a complete, legal chronological `PokerState` trajectory; `Scenario Mode` is a lossy study snapshot. Never merge their authorities or substitute nominal `facingSizeBb` for trusted incremental `callAmountBb` inside `DecisionContext v1 / v1.1`.
   - *Strategy & Explanation Authorities:* `DecisionContext v1` → `StrategyProvider v1` → `StrategyResult v1` is the sole strategy pipeline; `RangeAnalysisFacts v1` → `AnalysisExplanation v1` is the sole explanation pipeline. UI code formats and renders; it never computes poker math.
5. **The Player's Plan Is First-Class (Personal Strategy Three-Layer Model):** Personal Strategy is structured into three orthogonal layers (`RANGE-CAL-000` / `RANGE-CAL-001`):
   - **Profile:** A recognizable real-world game environment (e.g., *Home Game with Friends*, *ClubGG Freeroll*, *Online 9-Max Cash*).
   - **Mode / Anchor:** Up to **three user-named strategic approaches per profile** (e.g., *Normal*, *Cautious*, *Pressure*), treated as discrete anchors until empirical calibration data justifies continuous interpolation (`STRATEGY-FOUNDATION-002`).
   - **Spot Context:** Objective situational coordinates (table size, position, effective stack, facing action family, hand class).
   - *Elicitation & Provenance Rules:* A one-click calibration response records the **dominant/preferred action**, never silent 100% purity; exact mixed-frequency entry is always available as an optional refinement. Training sessions can contribute evidence via per-session opt-in (`training-memory/v1`), but conflicts between observed Training choices and deliberate calibration answers preserve both records and flag the deviation for user clarification rather than overwriting intent.
6. **Off-Table Study Workstation Only — Hard Anti-RTA Technical & Naming Controls:** The phrase *"Live Decision Mode"* is permanently retired across all UI, documentation, and store listings, replaced by **"Off-Table Scenario Simulator"** and **"Post-Session Spot Lab."** Riverline enforces five hard technical

boundaries: (a) zero overlays on poker clients, (b) zero screen reading or live hand-history tailing, (c) mandatory first-launch Anti-RTA Fair-Play click-through agreement, (d) automatic anonymization of player screennames and operator hand IDs (`Hero / BTN / BB`) on all shared hands, and (e) voluntary desktop process detection in Electron that pauses solver and chart views whenever known online poker clients (`PokerStars.exe`, `GGnet.exe`, `888poker.exe`, `WPN`) are running **[Judgement]**.

7. **Free First, Legally Audited, and Unitless Points-Only Scorekeeping:** Every external solver output pipeline, code library, and audio asset is recorded in the license registry before use. The **Game Night** organizer operates exclusively in **unitless Points / Chips-Only Mode** (*Points In*, *Points Out*, *Net Score Differential*, *Minimum-Step Chip Reconciliation*, *Host Tally*) with zero currency symbols (₪, $, €, ₽), zero debt/banker/IOU labels, zero rake/tip fields, and 100% local storage **[Judgement]**.

### 2.3 Technology, Design System, and Agent Working Rules

- **Technology Stack:** Plain JavaScript (`app/src/core/logic.js` ~14k lines + modular ES subsystems), single stylesheet `app/styles.css` owning the unified `:root` token block (`DS-FOUNDATION-001`), component catalog at `/__dev/component-sheet.html`, shared domain in `shared/poker-domain/`, ~2,750 Node DOM tests, `PERF-001` invocation-count guards, and bounded offline solver scripts in `solver/` isolated from production runtime.
- **Design System (`SHELL-001`, `SAVED-COMPOSITION-002`, `ANALYZE-COMPOSITION-001`):**
  - *Themes & Hierarchy:* **Midnight** (dark forest / jade / brass primary mockup palette), **Graphite**, and **Daylight**. One 56px header bar, one canvas grid (`--canvas-max: 1680px`, 24px gutter), one Spot rail + one Result panel in Analyze, one toolbar + sticky keyboard-navigable inspector in Saved, and `Explain` collapsed by default (`QA-ANALYZE-COMPOSITION-002`).
  - *Typography Scale:* Display `40px`, Title `20px`, Heading `15px`, Body `14px`, Small `12.5px`, Label `11px` (no product copy below `12px` except overlines).
  - *Motion & Foley Audio:* 31-moment motion catalog at 60 fps (`transform / opacity / clip-path` only; keyboard/frequent actions instant; global `Instant / Faster / Standard / Reduced` toggle). Audio uses 100% CC0 close-miked real recordings from **Set A + Set B** across 13 events, normalized to 16-bit / 44.1 kHz with a −3 dB high-shelf roll-off above 5.5 kHz.
- **How Any Agent Must Work on Riverline:**
  1. *Read Order:* Read this Extended Brief, `AGENTS.md`, `CURRENT_PHASE.md`, the relevant subsystem spec, and `QA_BACKLOG.md` before proposing or editing code.
  2. *One Bounded Outcome Per Ticket:* Inspect the canonical implementation first; never create duplicate poker, Equity, Training, strategy, or state authorities; leave work unstaged and uncommitted for human review unless instructed otherwise.
  3. *Evidence First in Research:* Open the primary source, quote the exact clause or file:line, attach clickable URLs, mark analytical inferences as **[Judgement]**, and mark uninspected items as **UNVERIFIED**.

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## SECTION 3 · TRACKS 1 & 2 — SOLVER LICENSING, PREFLOP/MULTIWAY ENGINE & VERIFICATION

## 3. Deep-Dive: Solver Intellectual Property Doctrine, Multiway Preflop Architecture, and Verification Tolerances

The strategic core of a serious poker workstation rests on the mathematical integrity and legal cleanliness of its reference baselines. Riverline's architectural evaluation separates solver usage into two distinct operational modes: **Mode 1 (Offline Data Generation / Pre-Computed Strategy Redistribution)**, where solver binaries run on private developer hardware to export static numerical arrays of action frequencies and expected values; and **Mode 2 (Embedded Engine / In-App Code Inclusion)**, where solver source code or binaries are linked into the distributed client or exposed over a network API **[Judgement]**.

### 3.1 Copyright, Copyleft (AGPL-3.0), and Contractual EULA Analysis

For Mode 1 offline data generation, open-source copyleft licenses and commercial EULAs diverge sharply in their legal mechanics:

- **Why AGPL-3.0 Does Not Infect Offline Numerical Strategy Outputs (b-inary/postflop-solver):** Under Section 2 of the GNU AGPL-3.0, the output of a covered program is subject to copyleft only if the output itself constitutes a covered work of code or creative assets. The Free Software Foundation's official guidance (FSF GPL FAQ #WhatCaseIsOutputGPL) states: "*The output of a program is not, in general, covered by the copyright on the code of the program. So the license of the code of the program does not apply to the output, whether you pipe it into a file, make a screenshot, screencast, or video.*" Furthermore, Nash equilibrium probabilities and expected values are uncopyrightable mathematical facts under 17 U.S.C. § 102(b) (*Feist Publications v. Rural Telephone*), and any EU/UK *sui generis* database right (Directive `96/9/EC`) vests in Riverline as the party investing the compute resources **[Judgement]**. Exporting floating-point arrays from `b-inary/postflop-solver` into Riverline's own clean serialization format (flatbuffers, SQLite, or JSON) is therefore 100% legal in closed-source free (Scenario a) and closed-source paid (Scenario b) distributions, provided zero Rust source code from `postflop-solver` is linked into the shipped client or hosted as a network solver service under AGPLv3 Section 13 **[Judgement]**.
- **Why HoldemResources Calculator (HRC Pro v4.0+) Is the Sole Commercial Engine Cleared for App Redistribution:** Unlike competing commercial desktop solvers, the HRC v4.0+ EULA (governed by Austrian law) contains an explicit commercial output grant:

  > *"Sharing the license key or making the Product accessible to others is prohibited, especially through automated means. This includes integrating the Product into the backend of a web service to allow others to submit calculations or use the Product indirectly. However, the Licensee is permitted to distribute, openly share, or use the results of authorized calculations for commercial purposes, provided these results were produced by the Licensee in compliance with this Agreement and applicable law... Automated scraping of information from the Product's user interface or memory is expressly prohibited. Results should only be exported using the export and save options provided within the Product."* — *HoldemResources Calculator v4.0+ EULA*

  At $359.90/year for HRC Pro ($29.99/mo billed annually or $119.97/quarter), a single developer license legally permits shipping 16,384-bucket multiway preflop, raked cash, ICM, and PKO calculation results inside a closed-source commercial app when exported via HRC's native export menu **[Judgement]**.
- **Why GTO+, Simple Postflop, PioSOLVER, GTO Wizard, and GTOpen Are Excluded:**

- GTO+ EULA: *"The license that is granted to You by the Company is solely for your personal, non-commercial purposes strictly in accordance with the terms of this Agreement."* Banned across all public app scenarios without a written waiver from StoxEV B.V.
- Simple Poker ToS § 2: *"You agree not to reproduce, duplicate, copy, sell, resell or exploit any portion of the Service, use of the Service... without express written permission by us."* Banned without a signed B2B contract.
- PioSOLVER Licensing Terms: *"you can't turn into on-demand service where specific PioSOLVER results would be available without actually purchasing PioSOLVER program."* Serving queryable PioSOLVER node frequencies on demand inside an app requires the €5,000/month PioSOLVER Business License (though Riverline may legally offer an optional local UPI socket bridge for users who already own a personal licensed copy of PioSOLVER).
- GTO Wizard API & Leaderboard Terms § 4.1: Explicitly forbids scraping or using benchmarking data *"to train, fine-tune, or calibrate any third-party artificial intelligence or poker solver (i.e., 'Model Distillation')."*
- MatthewPDingle/GTOpen: Contains no `LICENSE` file in its repository root. Under the Berne Convention and US Copyright Act, public repositories without a license remain **All Rights Reserved**. Do not copy code or ship outputs from `GTOpen` unless @MatthewPDingle commits an OSI-approved license file **[Judgement]**.

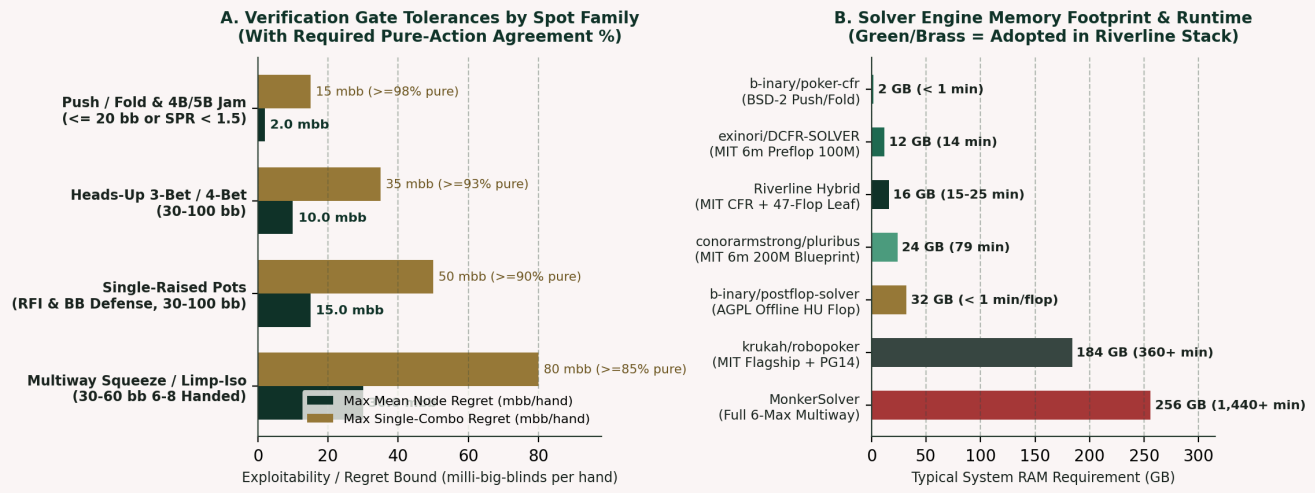



**Figure 2 — Verification Gate Regret Tolerances (ΔEV) and Solver Engine Resource Profiles [Judgement]:** Panel A specifies the maximum acceptable Mean Node Regret (mbb/hand), Max Single-Combo Regret, and minimum Pure-Action Agreement required to certify a preflop/postflop spot as `Verified`. Panel B compares system RAM footprint and convergence runtimes across candidate engines.

### 3.2 Complete Technical & Licensing Matrix for All 14 Evaluated Solvers & Frameworks

To translate the legal doctrine of Section 3.1 and the verification thresholds of Figure 2 into concrete engineering decisions, every candidate solver and game-theoretic library in the 2026 ecosystem was audited across five axes: game-tree coverage (player count, street depth, rake, and card bunching), empirical benchmark accuracy, hardware RAM/CPU footprint, verbatim license clauses, and legal shippability across closed- and open-source tiers **[Judgement]**.

| Solver / Framework | Game Coverage & Architecture | Benchmark Accuracy & Speed Evidence | Licence & Cost | Riverline Shippability Verdict [Judgement] |
|---|---|---|---|---|
| **b-inary/postflop-solver** (desktop / wasm) | HU postflop (Flop/Turn/River); raked/unraked; geometric sizes; **exact card bunching (up to 4 folded seats)**. | Rust DCFR (γ = 3.0), AVX2/NEON SIMD, zero card abstraction, exact isomorphism. Outperforms PioSOLVER & GTO+ on Ryzen 7 3700X at 0.1% dEV. | AGPL-3.0-or-later · $0 (Suspended Oct 2023, Issue #46). | **PRIMARY HU POSTFLOP ENGINE.** Offline outputs shippable in Closed/Open × Free/Paid. |
| **conorarmstrong/pluribus** (NoRegrets) | 2–6 players (extensible to 8–9 in `src/engine.rs`); External-Sampling Linear MCCFR + multiway flop search (`src/search.rs`). | 200M-iter blueprint matches real Pluribus top action **75.6% preflop** across 10,000 *Science* hands; HU BR bound **+116 mbb/h** (79 min on 16 cores, 16–32 GB RAM). | MIT License on public code ($0; pretrained `.bin` withheld). | **PRIMARY MULTIWAY & LBR ENGINE.** Train in-house; ship outputs & code in Closed/Open × Free/Paid. |
| **exinori/DCFR-SOLVER** | 6-max preflop (`src/preflop.rs` via External Sampling MCCFR, 30 SRP/3B/4B ranges) + HU postflop (DCFR/CFR+/QRE). | Postflop verified to **0.016% exploitability** (10K iters); preflop solves 2.2M infosets at 122K iters/s (**100M iters in 14 min**, <16 GB RAM). Leaf model **UNVERIFIED**. | MIT License · $0. | **PRIMARY FIXED-TREE PREFLOP ENGINE.** Pair with 47-flop `b-inary` leaf bridge; shippable everywhere. |
| **krukah/robopoker** | 2p & 6-max (`sixmax` flag) NLHE & Short Deck; Sinkhorn/EMD k-means++ + MCCFR + `litmus` 40-invariant suite. | Plays within **−13.1 ± 14.0 bb/100** vs Slumbot (86K hands); passes 34/40 GTO shape invariants in `litmus` (Flagship needs 120 GB RAM + PG14). | MIT License (12 crates on `crates.io`) · $0. | **ADOPT FOR `litmus` INVARIANTS & ACTION TRANSLATION.** Shippable in Closed/Open × Free/Paid. |

| Solver / Framework | Game Coverage & Architecture | Benchmark Accuracy & Speed Evidence | Licence & Cost | Riverline Shippability Verdict [Judgement] |
|---|---|---|---|---|
| **b-inary/poker-cfr** | HU exact Nash push/fold (`main_push_fold.rs`) & HU check-down preflop (`main_preflop.rs`). | Exact CFR+ over precomputed all-in equities (<0.001 bb exploitability; <4 GB RAM). | BSD-2-Clause · $0. | **ADOPT FOR SHORT-STACK PUSH/FOLD.** Extend to 6–9 seats with bunching; shippable everywhere. |
| **HRC Pro v4.0+** | 2–10p multiway preflop (ChipEV, raked cash, ICM, PKO) + **16,384 postflop realization buckets**. | Commercial gold standard for 6–9 max preflop & multiway realization; unlimited tree size. | Proprietary EULA · $359.90/yr ($29.99/mo). | **APPROVED PAID OPTION (M3).** Native exports explicitly cleared for commercial redistribution. |
| **MatthewPDingle/GTOpen** | Preflop Lab (2–9p, straddles, limps, 0/% rake) + HU postflop (CPU/CUDA). | Uses coupled-deck equity for 3+ way leaves; warns of card-removal errors on tight ranges. | **NO LICENSE FILE** (All Rights Reserved). | **DO NOT USE OR SHIP** until author commits an explicit FOSS license file. |
| **bupticybee/TexasSolver** | HU postflop only (NLHE & Short Deck); C++/Qt + GPU edition. | b-inary benchmark found divergent, unverifiable EV outputs vs Pio/GTO+ on `3betpotFAST`. | AGPL-3.0 + conflicting commercial README claims. | **SKIP.** Inferior accuracy to `b-inary/postflop-solver` and contradictory README terms. |
| **PioSOLVER v3.x**, **GTO+**, **Simple Postflop** | Commercial desktop solvers (HU postflop; Simple also has 3-Way & Preflop). | High benchmark quality, but legally restricted from third-party app database redistribution. | Pio Business: €5,000/mo; GTO+: $75 personal; Simple: $299+ B2B. | **DO NOT SHIP OUTPUTS.** (Optional local UPI bridge allowed only if end user owns personal PioSOLVER.) |
| **MonkerSolver** | Multiway NLHE & PLO preflop/postflop via MCCFR abstraction. | Requires 200+ GB RAM for full 6-max trees; noisier low-frequency convergence than HRC Pro. | Proprietary ToS · €499 one-time. | **DEFER / SKIP FOR NLHE.** Local outputs allowed by ToS, but inferior to HRC Pro & MIT stack. |

| Solver / Framework | Game Coverage & Architecture | Benchmark Accuracy & Speed Evidence | Licence & Cost | Riverline Shippability Verdict [Judgement] |
|---|---|---|---|---|
| **open_spiel**, **slumbot2019**, **pokerkit**, **pokerstove** | Reference C++/Python engines (2–10p ACPC, DCFR, Linear MCCFR, multiway hand eval & side pots). | Academic reference standards (Brown & Sandholm 2019; IEEE ToG 2025 for PokerKit). | Apache-2.0, MIT, BSD-3-Clause · $0. | **APPROVED REFERENCE PRIMITIVES.** Fully shippable in Closed/Open × Free/Paid. |
| **Poker-MCCFRM**, **poker_ai**, **rlcard**, **Deep-CFR** | Prototype C#/C++/Python MCCFR and neural Deep CFR repositories. | `poker_ai` defaults to 20-card deck; `rlcard` has coarse 5-action space; Deep-CFR has high variance vs tabular 169-hand preflop MCCFR. | MIT (`rlcard`, `Deep-CFR`, `MCCFRM`) / GPL-3.0 (`poker_ai`). | **SKIP FOR PRODUCTION.** Outclassed by Rust tabular MCCFR crates (`pluribus` / `DCFR-SOLVER`). |

### 3.3 Engineering the Zero-Cost Preflop & Multiway Stack and Three-Tier Verification Gate

To fill the owner's preflop and multiway gap (6–8 handed 30–60 bb zero-rake home games and 9-max 100 bb+ raked online games) without spending money or waiting 18–36 months to write a full abstraction solver from scratch, Riverline will execute a three-stage engineering pipeline **[Judgement]**:

1. **Stage 1 — Exact Short-Stack Push/Fold & 3-Bet/4-Bet Jam Oracle (Weeks 1–2):** Extend b-inary/poker-cfr (BSD-2-Clause) and conorarmstrong/pluribus (MIT) to 6–9 seats for ≤ 15 bb open-shove/fold and 20–35 bb 3-bet/4-bet jam trees, incorporating folded-seat card-bunching weights. Certify outputs as **`Verified · Equilibrium`** (ΔEV ≤ 2.0 mbb/hand, ≥ 98% pure-action agreement).
2. **Stage 2 — Fixed-Size Preflop MCCFR + Offline 47-Flop `b-inary` Leaf Bridge (Weeks 3–6):** Fork exinori/DCFR-SOLVER and conorarmstrong/pluribus (MIT) to run External-Sampling Linear MCCFR (Brown & Sandholm, AAAI 2019; Lanctot et al., NeurIPS 2009) over 6–9 seat preflop trees with configurable rake (0 for Home Game, `0.1 bb`/seat for ClubGG, % capped for 100 bb online). Terminate the tree when preflop action closes and evaluate heads-up postflop leaves (SRP, 3-bet, 4-bet, BvB) using an Equity Realization table $R(h_i, \text{pos}_i, \text{SPR}, k)$ fitted across 200–500 offline b-inary/postflop-solver runs on a canonical 47-flop subset (184 flops for final release sign-off). Use pseudo-harmonic action translation (krukah/robopoker) to map off-tree live bet sizes. Certify outputs that pass the LBR/BR + `litmus` gate (ΔEV ≤ 10–15 mbb/hand, ≥ 90–93% pure-action agreement) as **`Verified · Equilibrium`**.
3. **Stage 3 — Restricted Nash Response (`--rnr-model`) for 30–60 bb Home-Game Limped Pots & 3-Way Flops (Weeks 7–12):** Because UTG–CO open-limping is a 0%-frequency off-tree action in equilibrium, solve 30–60 bb home-game isolation-raise and overlimp ranges via Restricted Nash Response in conorarmstrong/pluribus against 2–3 transparent recreational limper profiles (e.g., *Capped Loose-Passive Limper: VPIP 40% / PFR 8%, traps top 2%*) and label them **`Verified · vs. Loose-Passive Limp Model`** (ΔEV ≤ 30.0 mbb/hand, ≥ 85% pure-action agreement). For 3-way flops,

use 1-street depth-limited flop resolving (`src/flop.rs`) with 4 continuation strategies and coarse action guidance (`Check / Bet 33% / Fold`).

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SECTION 4 · TRACK 3 — BRAND, TRADEMARK, DOMAIN & MULTILINGUAL NAMING AUDIT

## 4. Brand Clearance Audit (EN · RU · HE) and Vetted Alternative Name Portfolio

The brand and naming investigation audited trademark registries (USPTO, India/WIPO Class 42, Rospatent, and public indexes for EUIPO, UK IPO, and the Israel Patent Office), domain registrars, Apple App Store and Google Play listings, developer namespaces, and linguistic connotations in English, Russian («Риверлайн»), and Hebrew («ריברליין»). The audit returned a verdict of **HIGH RISK [Judgement]** for retaining the name **Riverline** on any public poker study release.

### 4.1 Detailed Evidence Table of Collisions Against "Riverline"

| Audit Category | Status & Risk | Primary Evidence, Registration Numbers & Direct URLs |
|---|---|---|
| **1. Online Poker & Gambling Vertical** | **Direct Collision · HIGH (Critical)** | Riverline Poker (`riverlinepoker.com`) is a live, USDC-on-Polygon real-money online poker room and operator network offering NLHE cash games, fast-fold ("Quick Play"), SNGs, and tournaments under the exact brand **"Riverline"** (© 2026 Riverline), localized in **English, Spanish, and Russian** (House Rules v1.1, Sept 18, 2026; LMIS Review). Directly threatens Principle 6 and triggers Apple/Google gambling-front flags **[Judgement]**. |
| **2. Class 42 Software Trademark & Google Play App** | **Taken / Pending · HIGH** | Venture-backed fintech startup Riverline AI (Pointfront Technologies Pvt. Ltd., funded by $825K pre-seed led by South Park Commons) holds pending Class 42 trademark App. #6935248 ("Riverline") covering *"Smartphone Software Design, SaaS, and Third-Party Game Software Publishing"*, operates riverline.io, and publishes Riverline: Manage My Loans (`id=ai.riverline.app`, including Russian Play Store) under the legal header *"RIVERLINE APP"* (App Privacy Policy). Other active tech firms: Riverline Systems and Riverline Dubai. |
| **3. USPTO, UK & Global Trademarks** | **Registered in Adj. Classes · MEDIUM** | • USPTO Reg. #7428542 / Serial #98044162 (*RIVERLINE*, Brilliant Initiatives LLC, Class 11 HVAC/water filters, LegalHoop).<br>• USPTO Serial #78843317 (Riverline Ergonomic, Class 9 protective safety footwear, Riverline Work USA).<br>• USPTO Serial #76176881 (*RIVER LINE*, NJ Transit, Class 39).<br>• UK Companies House active entities: RIVERLINE LIMITED (01756395) and RIVERLINE TRADING LIMITED (06846691). Live interactive searches inside gated portals (EUIPO, UK IPO, ILPO, WIPO) are marked **UNVERIFIED**. |

| Audit Category | Status & Risk | Primary Evidence, Registration Numbers & Direct URLs |
|---|---|---|
| **4. Primary Domains & Developer Handles** | **Taken / Blocked · HIGH** | • riverline.com: Owned by NJ TRANSIT Light Rail since 2004.<br>• riverline.io & riverline.ai: Active portals for Pointfront Technologies (Official Guide).<br>• `riverline.app`: Reverse-DNS root of `ai.riverline.app`.<br>• github.com/Riverline: Occupied by a 13-repository organization.<br>• Apple App Store: Occupied by *River Line Schedule* (NJ Transit app).<br>• Social/Video: Occupied by @riverlinework, @riverlineproducts, Riverline YouTube, and nu-metal band Riverline on YouTube Music (Reddit). |
| **5. Russian («Риверлайн») & Hebrew («ריברליין»)** | **Hijacked / Cluttered · MED–HIGH** | • **Russian:** Search engines map «Риверлайн» to Revyline (RU TM #776368, oral-care leader with Revyline Club App), plus ООО «РИВЕРЛАЙН», West Garden Riverline, York Towers Tbilisi, and sportsbook/stable echoes (Winline Media Poker, Maxline Poker, GreenLine Poker).<br>• **Hebrew:** Unvocalized ריבר matches River Noodle Bar (Rishon LeZion, Ramat Gan) and rebar (Google Play); ליין echoes party promoter lines or betting odds lines, alongside Poker Lines on Google Play IL. |

### 4.2 Vetted Low-Risk Replacement Names Across English, Russian, and Hebrew

After disqualifying taken lookalikes—specifically FeltDesk, FeltBook, FeltFolio, RiverNote, RiverDesk, Riverframe (taken by an indie software studio at `riverframe.dev`), OrbitDesk (Google Play), OrbitNote, Rangebook, Clearfelt, and TurnBench—twelve single-word candidates passed web, `.com`/`.app`, app-store, and trilingual phonetic checks with **Low Risk [Judgement]**:

| Candidate (EN · RU · HE) | Poker Workstation & Aesthetic Resonance [Judgement] | Domain, App-Store & Search Clutter Audit |
|---|---|---|
| **1. Rangedesk**<br>«Рейндждеск» · «ריינג'דסק» | Unites **Range** (the core analytical unit of NLHE and Personal Strategy) with **Desk** (calm local-first study workstation). Signals serious off-table study rather than gambling or RTA. | `rangedesk.com` & `.app` unindexed; zero software, poker, or app-store hits; only 2-word furniture pages from UK retailer The Range. |
| **2. Turndesk**<br>«Тёрндеск» · «טרנדסק» | Pairs **Turn** (4th street where pot geometry sharpens + "your turn to act") with **Desk**. Crisp 2-syllable phonetics across EN, RU, and HE with zero negative slang. | `turndesk.com` & `.app` unindexed; zero exact matches on web or app stores (engines auto-suggest vinyl Turntable Lab). |
| **3. Riverlamp**<br>«Риверлампᴨ» · «ריברלמפ» | Retains **River** and pairs it with **Lamp**—evoking a classic green-shaded brass banker's lamp illuminating the table (*"dark forest / jade / brass"*, *"Every number says where it comes from"*). | `riverlamp.com` & `.app` unindexed; zero software, gaming, or trademark hits; only generic 2-word decor listings (Etsy River Lamp). |

| Candidate (EN · RU · HE) | Poker Workstation & Aesthetic Resonance [Judgement] | Domain, App-Store & Search Clutter Audit |
|---|---|---|
| **4. Handbench** «Хэндбенч» · «הנדבנץ'» | Combines **Hand** (*"for the hands you actually play"*) with **Bench** (both a craftsperson's workbench for hand reconstruction and a reference benchmark for `Verified` tiers). | `handbench.com` & `.app` unindexed; zero software or poker hits; only industrial screen-printing tables (Wicked Printing Stuff). |
| **5. Rangebench** «Рейнджбенч» · «ריינג'בנץ» | Unites **Range** with **Bench**, fitting a workstation that benchmarks Personal Strategy against a reference baseline without calling deviations "mistakes." | `rangebench.com` & `.app` unindexed; zero software/app hits; only 2-word furniture item (Fort Standard Range Bench). |
| **6. Handfolio** «Хэндфолио» · «הנדפוליו» | Pairs **Hand** with **Folio** (a structured portfolio/notebook of reconstructed hands, reviews, and strategy plans). Calm, editorial tone in EN/RU/HE. | `handfolio.com` & `.app` unindexed; zero exact matches in web search or app stores (falls back to leather padfolios like Levenger). |
| **7. Rangefolio** «Рейнджфолио» · «ריינג'פוליו» | Pairs **Range** with **Folio**, emphasizing a living personal strategy book and range library owned by the player. | `rangefolio.com` & `.app` unindexed; zero software/app matches (falls back to AT-A-GLANCE padfolios). |
| **8. Riverstudy** «Риверстади» · «ריברסטאדי» | Keeps **River** while explicitly appending **Study**, eliminating any ambiguity with Riverline Poker or RTA for Apple/Google reviewers. | `riverstudy.com` & `.app` unindexed; zero software/poker hits; only environmental pages (Fox River Study Group). |
| **9. Turnline** «Тёрнлайн» · «טרנליין» | Closest structural twin to *Riverline*, shifting to 4th street (**Turn** + betting **Line**) where postflop branching and study complexity peak. | USPTO #75333950 (*TURNLINE* helmets) Dead since 2007 (Trademarkia); only a lathe sub-model (Tungaloy); `.com`/`.app`/`.io` unindexed. |
| **10. Rangeframe** «Рейнджфрейм» · «ריינג'פריים» | Doubles as a poker **Range** framework and Edward Tufte's classic statistical **range-frame** plot axis—ideal for an honest data workstation. | Only appears as an academic plotting function in MATLAB/R (MathWorks); `rangeframe.com` & `.app` unindexed; zero apps. |
| **11. Feltlog** «Фелтлог» · «פלטלוג» | Table **Felt** + study **Log**: tactile, calm, 2-syllable name signaling a quiet off-table hand journal rather than casino action. | `feltlog.com` & `.app` unindexed; zero mobile apps or trademarks (avoid *Feltnote*, used by EthNote). |
| **12. Flopframe** «Флопфрейм» · «פלופפריים» | Strong F–F alliteration (**Flop** + **Frame**); immediately recognizable to any Hold'em player with 1-to-1 spelling in EN, RU, and HE. | Completely clean: zero web results, zero apps, zero trademarks, and no indexed sites on `flopframe.com` or `flopframe.app`. |

SECTION 5 · TRACK 4 — COMPETITIVE LANDSCAPE & INTERACTIVE LEARNING UX BENCHMARKS

## 5. Competitive & Pedagogical Survey Across 12 Poker Product Families and 2 EdTech Leaders

The 2026 poker study software market is sharply bifurcated between expensive cloud solver platforms built for high-volume online grinders (GTO Wizard, GTOBase) and fragmented mobile quiz apps that

reduce multi-street range interactions into binary right/wrong flashcards (DTO Poker, Postflop+) **[Judgement]**. Synthesizing user feedback across TwoPlusTwo Live No-Limit Cash forums, r/poker, r/Poker_Theory, r/chess, and r/math reveals why live and home-game players remain systematically underserved—and how Riverline can capture that white space.

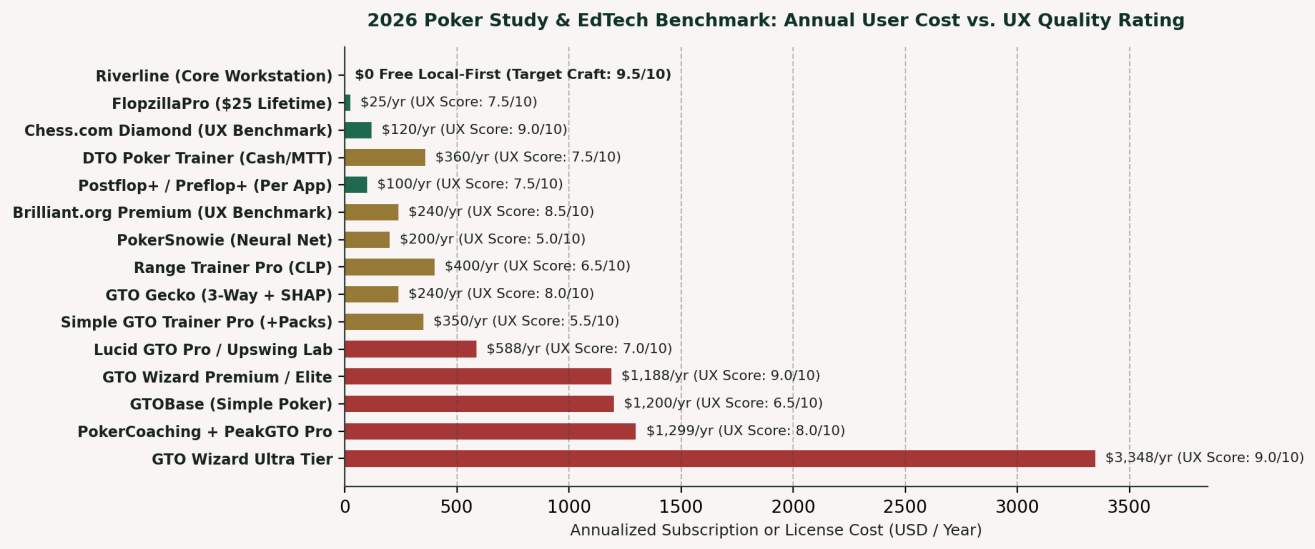



**Figure 3 — 2026 Poker Study & EdTech Annual Cost vs. UX Quality Rating [Judgement]:** Annualized subscription or license cost (USD/year) across 14 benchmarked products compared against Riverline's $0 local-first model. Commercial cloud trainers charge $468 to $3,348 per year while failing to support zero-rake multi-limp home games.

## 5.1 Master Comparison Matrix of 14 Poker & EdTech Products (2026)

| Product Family | What Users Love | What Users Complain About (Forums / Reviews) | Pricing (2026) | Uncertainty & Live/Home Fit [Judgement] |
|---|---|---|---|---|
| **GTO Wizard** (Best Apps 2026) | 10M+ presolved library, fast cloud AI solving, nodelocking, instant Trainer-to-Study matrix toggle. | 5-tier "rentalism" (r/poker WTF thread), cognitive overload, unrealistic HU LAG assumptions that hurt live winrates (TwoPlusTwo). | Free (1 spot/day); **$39–$279/mo** ($468–$3,348/yr). | Treats solver equilibrium as exact truth; blocks multi-limp family pots and lacks recurring human profiles. (UX: **9.0/10**) |
| **GTOBase** (Simple Poker) | Fast web strategy viewer, generous free Spin & Go / HU libraries (Alternatives). | iOS app unupdated since July 2023 (v2.09), post-payment login bugs, zero explanation of bizarre lines (e.g., UTG calling 3B OOP with 33). | Free subset; **$75–$150/mo** ($1,200/yr) per library. | Raw Simple Postflop numbers with zero "why" explanation and zero live/home-game support. (UX: **6.5/10**) |

| Product Family | What Users Love | What Users Complain About (Forums / Reviews) | Pricing (2026) | Uncertainty & Live/Home Fit [Judgement] |
|---|---|---|---|---|
| **DTO Poker Trainer** (Cash & MTT) | Low-friction one-handed mobile swipe loop; $9.99/mo entry tier. | **Teaches false purity:** grades mixed spots by rewarding only the modal action (r/poker Flop Thread); split into separate Cash & MTT apps. | Freemium; **$9.99–$99.99/mo**. | Masks mixed frequencies in lower tiers; zero live sizing, multiway, or home-game support. (UX: **7.5/10**) |
| **Postflop+ & Preflop+** | True offline downloadable spot packs, $199.99 lifetime option, Preflop+ pot-odds & combinatorics drills. | **Severe 5-app fragmentation** (Preflop+, Postflop+, Solver+, Ranges+ billed separately); no range explorer inside Postflop+ drills. | **$14.99/mo**, **$99.99/yr**, or **$199.99 lifetime** per app. | Shows street-by-street range narrowing, but lacks conceptual explanations and multiway postflop. (UX: **7.5/10**) |
| **Simple GTO Trainer** | Lets power users import and drill custom local Simple Postflop or PioSOLVER files. | Windows-only; requires buying a $299–€800 solver or $59–$350 packs on top; archaic 1-PC Skype license deactivation. | **$199/yr** Pro + **$59–$349.99** per pack. | Exact EV loss vs imported file; out-of-the-box packs are 100% online trees. (UX: **5.5/10**) |
| **Range Trainer Pro** (CLP) | Active-recall flashcard drilling of custom preflop ranges; praised by live pros for warmups (Jim Reid). | $30–$37/mo just for preflop chart memorization (r/poker Preflop); rote memorization of 50+ charts without principles (NLH Thread). | **~$30–$37/mo** subscription. | Pure binary chart matching (Range Charts); supports custom live preflop charts (CPG), preflop only. (UX: **6.5/10**) |
| **FlopzillaPro** (Purchase) | Legendary **$25 lifetime** license (2 PCs); instant visual range-vs-board hit breakdown, 3-way equity, and bunching. | Windows-only binary; dense legacy toggle UI; minor top-% hand ordering differences vs Equilab (TwoPlusTwo; Bug Thread). | **$25 one-time lifetime** (zero subscriptions). | **100% deterministic combinatorial & equity math**—never invents fake strategy verdicts; great for manual live ranges. (UX: **7.5/10**) |
| **PokerSnowie** (Review 2026) | Gentle beginner play-and-grade loop; candid public AI Weaknesses page. | 2013 neural net uses only 4 bet sizes, forces 1 size per range, folds standard EP opens, fails vs limpers/overbets; mobile apps dead since 2019. | **$29.90/mo** or **$199.90/yr**. | Black-box neural "Error Rate" makes documented pot-odds blunders in all-in and limped pots (AI Weaknesses). (UX: **5.0/10**) |

| Product Family | What Users Love | What Users Complain About (Forums / Reviews) | Pricing (2026) | Uncertainty & Live/Home Fit [Judgement] |
|---|---|---|---|---|
| **Upswing Lab 2.0 & Lucid GTO** | Beginner "Four-Category System", 28-hr *Live Cash Game Bankroll Booster* track, flat Lucid pricing + free Basic tier. | Overwhelmingly passive video LMS with near-zero interactive exercises (r/Poker_Theory); $999 course upsell fatigue (Modules Thread). | Lab **~$99/mo**; Lucid Free or **$49/mo** Pro. | Strong live cash video curriculum, but disconnected from interactive personal range building. (UX: **7.0/10**) |
| **Run It Once & Vision** | Elite pro instruction (Phil Galfond, Peter Clarke), **gold-standard provenance transparency in Vision FAQ**, Vision exploit drills (Hunter Cichy). | Unstructured library outside learning paths (RIO vs Upswing); passive video lectures. | Videos **~$100/mo** ($49 FTGU); separate Vision plan. | Explicitly discloses Monker buckets `(30,30,30)`, `I/N` volatility, rake, and human sizing simplifications (Vision FAQ). (UX: **8.0/10**) |
| **HM3 & Hand2Note 4** | H2N imports 10k hands/s, embeds graphical hands inside notes, attaches notes to specific stats, and filters out "Reg-vs-Fish" hands (H2N Review). | Steep setup; HM3 sits at 2.2/5 on Trustpilot; HUDs banned/restricted on major sites; requires online HH text files. | HM3 **$60–$160**; H2N Free Base or **$19.90–$49/mo**. | Empirical sample stats without small-sample noise warnings; useless for physical live/home games without manual reconstruction. (UX: **H2N 8.0 / HM3 6.0**) |
| **PokerCoaching / GTO Gecko / Live Poker Trainer** | GTO Gecko **SHAP plain-English factor weights**; Live Poker Trainer **4 archetypes (Fish/Nit/Maniac/GTO)** + Easy/Med/Hard sizing buckets; Coach 0–10 quizzes. | PokerCoaching criticized as "wildly expensive" ($147/mo or $999–$1,299/yr); mobile apps track device IDs. | PC **$999–$1,299/yr**; Gecko **$24.99–$39.99/mo**; LPT **$19.99/mo**. | Best live focus among mobile apps (3-way sims + Fish/Nit/Maniac exploit drills), though behind heavy paywalls. (UX: **8.0/10**) |
| **Chess.com Game Review** | **Macro Advantage Graph → Key Moves → 1-click Retry** before reveal → visual hint arrows → direct opening lesson link (Membership). | Paywalling Stockfish (r/chess); shallow AI Coach text; depth whiplash; manipulative CAPS2 50–95 accuracy curve & rating-inflated "Brilliant!!" badges. | Free (1/day); **$4.17–$16.99/mo**. | Adopt the *Key Moments → Retry → Lesson* funnel; strictly reject CAPS2 score curving and dopamine inflation. (UX: **9.0/10**) |

| Product Family | What Users Love | What Users Complain About (Forums / Reviews) | Pricing (2026) | Uncertainty & Live/Home Fit [Judgement] |
|---|---|---|---|---|
| **Brilliant.org** (Review) | Draggable visual manipulatives (*Concrete → Pictorial → Abstract*), Socratic AI tutor Koji, zero passive video lectures. | "Edutainment" illusion of mastery without sufficient drill volume (r/math); abrupt difficulty cliffs; coercive **2-keys/day linear lock**. | Free (2 keys/day); **$20–$30/mo** ($240/yr upfront). | Adopt interactive visual manipulatives for range/pot geometry; reject daily key locks and forced linear topic gates. (UX: **8.5/10**) |

### 5.2 Addressing the Seven Unmet Needs of Home-Game & Live Players

The competitive synthesis identified seven acute gaps experienced by live and home-game players (TwoPlusTwo Live Cash; r/poker Live Thread; GTO Apps Pricing Thread) that Riverline's architecture solves natively **[Judgement]**:

1. **Limp-Family & Isolation-Raise Pots:** Standard solvers reject multi-limp sequences (e.g., UTG limps, MP overlimps, CO isolates to 6 bb, called 3-way). Riverline supports arbitrary legal multi-limp states in `Hand Mode`, `Analyze`, and `Equity`, and models 30–60 bb limped preflop strategy via `Verified · vs. Model` (Restricted Nash Response).
2. **Partial-Memory Post-Session Hand Reconstruction:** Live players rarely remember all 8 folded seats' exact preflop stacks. Riverline's strict separation between **canonical `Hand Mode`** (full legal trajectory, side pots, and replay) and **lossy `Scenario Mode`** (fast study snapshot where missing history stays `unknown`) lets players reconstruct hands honestly within 30 seconds.
3. **Persistent Profiles for Recurring Human Opponents:** Unlike anonymous 5,000-player online pools, a weekly home game features the same 8–12 friends every week. Riverline's user-named `Profiles` × `3 Named Modes` plus `Opponent Policy v2` let players calibrate concrete plans for their actual table.
4. **Guarding Against "Piecemeal GTO" Traps:** Firing a 33% flop range c-bet in a loose-passive home game without knowing how to barrel turn overbets loses money against calling stations (TwoPlusTwo). Riverline's multi-street `PERSONAL-STRATEGY-CONTINUATION-001C` and `Bluff / Exploit Teacher v1` surface street-to-street plan coherence and fold-equity thresholds explicitly.
5. **Non-Casino Accounting:** Native support for **zero-rake Home Games (2–10 seats)** and **ClubGG-style `0.1 bb`/seat forced contributions (7–10 seats)** prevents 5% casino rake assumptions from distorting preflop flatting ranges.
6. **Zero Subscription Clock for Occasional Volume:** A player who plays 15 hours of home-game poker per month loses $6–$18 per hour of table time if paying $99–$279/month for a cloud solver. Riverline is local-first and free.
7. **Unified Points-Only Game Night Tally + Saved Hand Bookmarking:** Hosts can tally session points in Game Night (unitless Points-Only Mode) while bookmarking key showdowns directly into `SavedStudyObject v1` for post-game replay and group discussion.

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**SECTION 6 · TRACK 5 — AUDIO LICENSING AUDIT & 52-CANDIDATE FOLEY CATALOG**

## 6. Acoustic Foley Source Licence Audit and 13-Event Sound Design Catalog

To fulfill Section 4 of the brief (*"Sound: only real, properly licensed recordings, judged by ear; study cues never sound like right/wrong"*), nine audio libraries were audited for commercial viability in a closed- or open-source application with a paid tier. Crucially, the audit flagged and excluded BBC Sound Effects (RemArc Licence), whose Clauses 4(f), 7, and 8(a) restrict free usage strictly to *"non-commercial, personal or research purposes"* and require paid commercial clearance via Pro Sound Effects for any product with a commercial or revenue-generating tier **[Judgement]**.

### 6.1 Complete 13-Event Real-Recorded Foley Catalog (52 Verified Candidates Across Sets A–D)

All 52 candidates below are real physical recordings (clay poker chips, plastic-coated playing cards, wooden tables, pistachio shells, orchestral bell trees, and Tibetan singing bowls—zero synthesized arcade bleeps) grouped into four acoustically coherent sets: **Set A (Dry Studio Felt & Wood, 100% CC0)**, **Set B (Close-Miked Clay & Card Room on Blue Yeti / Zoom H2n, 100% CC0)**, **Set C (Natural Acoustic Wood Table on Zoom H4n/H5n / TASCAM DR-40X, CC-BY 3.0/4.0 + CC0)**, and **Set D (Commercial Studio Foley Booth, Compiled-App Royalty-Free)**. Combining **Set A + Set B** provides a 100% CC0, zero-attribution production soundscape when exported at 16-bit / 44.1 kHz with a gentle −3 dB high-shelf roll-off above 5.5 kHz **[Judgement]**:

| Event # & Name | Set A Candidate (Dry Studio · 100% CC0) | Set B Candidate (Close Clay · 100% CC0) | Set C Candidate (Wood Room · CC-BY/CC0) | Set D Candidate (Compiled App Only) |
|---|---|---|---|---|
| **1. Card Dealt / Slid** | cardSlide1–8.ogg (Kenney, CC0, 0.40–0.50s, 44.1kHz Stereo) — 8 smooth felt slides. | slideCard04 (817579) (SilverDubloons, CC0, 0.35s Blue Yeti) & Using playing cards (749323) (Magnesus, CC0, 48kHz Zoom H2n). | place_cards.wav (45816) (themfish, CC-BY 4.0, 0.80s, 48kHz Stereo). | Single card remove & slide 6 (ZapSplat, 1.0s, credit on Free tier). |
| **2. Board Card Placed** | cardPlace1–4.ogg (Kenney, CC0, 0.30–0.50s) & contact1–2.wav (BMacZero, CC0, 0.10–0.30s crisp contact). | Using playing cards (749323) (Magnesus, CC0, 0.35–0.55s individual placement slices). | place_cards.wav (45816) (themfish, CC-BY 4.0, 0.80s natural wood table). | Playing card place down 1 & 2 (ZapSplat, 0.40s dry booth foley). |
| **3. Card Flip** | cut.wav (BMacZero, CC0, 0.20s, 44.1kHz WAV — quick single-card turn snap). | Index Card Flip (339015) (ROBAMOS, CC0, 0.45s close-miked cardstock flick). | Card Flip (240776) (f4ngy, CC-BY 4.0, 0.589s natural home turn-over). | Poker card flick (Mixkit License, 1.0s studio corner flick). |
| **4. Check Knock on Felt** | dedotoca mesa (563818) (Churrumaki, CC0, 0.45s damped finger/knuckle double-tap on table). | Knocking on Wood (629987) (Flem0527, CC0, 0.704s w/ 2kHz LPF) & Tapping Objects (863890) (firespectrum, CC0). | Plastic Tap (846553) & Wood Knock (Cpfcfan10 Pack 45251, CC-BY 4.0, TASCAM DR-40X). | Negative tone interface tap (Mixkit, 1.0s understated percussive thud). |

| Event # & Name | Set A Candidate (Dry Studio · 100% CC0) | Set B Candidate (Close Clay · 100% CC0) | Set C Candidate (Wood Room · CC-BY/CC0) | Set D Candidate (Compiled App Only) |
|---|---|---|---|---|
| **5. Chips Bet (S/M/L Stacks)** | chipLay1–3, chipsCollide1–4, chipsStack1–6 (Kenney, CC0, 13-file matched progression 0.25–0.55s). | chip05 (817556) (SilverDubloons, CC0, 0.216s real clay chip) & poker_chips6 (201808) (farthestar, CC0, 0.65s heavy clay). | Poker Chips on Wooden Table (445141) (Benjamin152, CC-BY 3.0, Zoom H4n 24-bit S/M/L drops). | Sonniss #GameAudioGDC Casino Foley (Sonniss License, royalty-free compiled). |
| **6. All-In Push** | chipsHandle5–6.ogg (Kenney, CC0, 0.75–0.90s heavy multi-stack slide and cascade). | clatter10 (817567) (SilverDubloons, CC0, 0.80s clay handful shove) & Poker Chip Drop (Za-Games, CC0, 0.708s). | Heavy Stack Slice (445141) (Benjamin152, CC-BY 3.0, 1.40s weighty multi-stack collapse). | Mixkit Casino Stack Push (Mixkit, 44.1kHz WAV). |
| **7. Pot Pushed to Winner** | chipsHandle1–4.ogg (Kenney, CC0, 0.60–0.85s sweeping gather across felt). | Handfull of Poker Chips (Za-Games, CC0, 0.352s warm clay scoop and gather). | Gather/Sweep Slice (445141) (Benjamin152, CC-BY 3.0, 1.20s clay sweep on wood). | Coins/Tokens handling (Mixkit, 1.0s w/ 4kHz high-shelf softening). |
| **8. Fold (Cards Tossed)** | cardShove1–4.ogg (Kenney, CC0, 0.40–0.55s two-card hand tossed into muck). | Playing Card Slide Right (571581) (el_boss, CC0, 0.45s airy lateral muck slide). | slap-cards.wav (45821) (themfish, CC-BY 4.0, 0.93s folded hand dropped on wood). | Handful of cards drop 5 (ZapSplat, 1.0s loose card toss). |
| **9. Soft UI Tick** | Pistachio_Percussion_17 (868796) (SignatureSoundsOrg, CC0, 96kHz, 0.08s organic shell micro-click). | Click_9 (872341) (CC0, 0.47s) & Radial knob clicks (643563) (el_boss, CC0, 0.12s rotary detent). | Glass Tap (846554) (Cpfcfan10, CC-BY 4.0, 0.159s delicate finger tap on glass). | Typewriter soft click (Mixkit, 0.09s muted mechanical tap). |
| **10. Confirm** | wood block.wav (53403) (calaudio, CC0, 0.433s warm acoustic woodblock) & Pistachio_Perc_19 (869735) (CC0, 0.18s). | billiard ball clack (539854) (Za-Games, CC0, 0.239s solid phenolic-resin clack). | Button Click (858919) (DeWildAshDev, CC0, 0.847s physical household foley switch). | Mixkit Tactile Confirm (Mixkit, 44.1kHz WAV). |
| **11. Gentle Error** | open-cardboard-box (Pack 20516) (newagesoup, CC0, 0.25s hollow card-case thud, zero buzzer). | Pistachio_Perc_25 (869741) (SignatureSoundsOrg, CC0, 0.22s lower-pitched damped shell double-tap). | Plastic Tap (846553) (Cpfcfan10, CC-BY 4.0, 0.60s dull non-resonant tap). | Negative tone interface tap (Mixkit, 1.0s low drum-skin tap). |

| Event # & Name | Set A Candidate (Dry Studio · 100% CC0) | Set B Candidate (Close Clay · 100% CC0) | Set C Candidate (Wood Room · CC-BY/CC0) | Set D Candidate (Compiled App Only) |
|---|---|---|---|---|
| **12. Session Finish** | bell_tree_scrape3 (375598) (sgossner, CC0, 1.8s SM-57 orchestral bell-tree brush) & shuffleandbridge.wav (BMacZero, CC0, 1.8s). | Singing Bowl (521549) (BuddhasSoundOfSilence, CC0, 2.5s warm Tibetan singing-bowl mallet glow). | Short vibraphone melody (753347) (xkeril, CC0, 2.8s Zoom H5n acoustic vibraphone chord). | Sonniss Warm Acoustic Finish (Sonniss GDC Royalty-Free). |
| **13. Toast / Notification** | chimes_part_6.wav (50405) (dADDoiT, CC0, 0.80s wooden clapper touching single chime tube). | bell ding 1.wav (611113) (5ro4, CC0, Blue Yeti desk bell w/ 3.5kHz LPF) & Small Hour Glass (625655) (el_boss, CC0, 0.90s sand whisper). | bell.wav (868263) (Mihacappy, CC0, 0.95s natural room acoustic bell chime). | ZapSplat Subtle Wooden Chime (ZapSplat Standard License). |

SECTION 7 · TRACK 6 — MULTI-JURISDICTIONAL LEGAL, APP STORE & ANTI-RTA COMPLIANCE

## 7. Regulatory, Platform-Policy, and Anti-RTA Compliance Framework (Israel, Russia, EU, US)

Operating a poker software application from Israel across English, Russian, and Hebrew markets while targeting web browsers, desktop Electron builds, and mobile app stores intersects four high-enforcement regulatory regimes: (1) Israeli criminal and contract law governing prohibited games, privacy, and consumer subscriptions; (2) Apple App Store and Google Play real-money gambling and illegal gambling-aid prohibitions; (3) online poker operators' security and ecology rules against Real-Time Assistance (RTA); and (4) Russian Federal Communications Service (Roskomnadzor) extrajudicial domain blocking and data-localization rules **[Judgement]**.

### 7.1 Statutory & Platform Rules Driving Riverline's Three Mandatory Compliance Redesigns

- **Redesign 1 — Why Game Night Must Be Strictly Unitless Points-Only:** Under Chapter 12 of the Israeli Penal Law, 5737-1977 (Sections 224–235), organizing a "prohibited game" carries up to 3 years' imprisonment (Section 225) and participating carries up to 1 year (Section 226). Although the Supreme Court of Israel noted in *obiter dictum* in a foreign tournament tax appeal (C.A. 476/17 *Rafi Amit v. Assessing Officer Tel Aviv 4*) that tournament poker involves substantial skill, Israeli courts continue to classify domestic cash-game poker as a prohibited game unless it satisfies all three cumulative conditions of the narrow social exemption in Section 230 of the Penal Law: *(1) intended for a particular circle of persons, (2) does not exceed the scope of amusement or small-scale entertainment, and (3) is not held in a place of prohibited games*. Crucially, Section 233(3)(a) of the Penal Law establishes a statutory presumption that any premises where a prohibited game is played at least twice within six months is an illegal gambling venue ("מקום משחקים אסורים"), and Section 32(a) of the Contracts (General Part) Law, 5733-1973 renders gambling debts civilly unenforceable. Simultaneously, Google Play's Real-Money Gambling Policy bans apps that provide *"companion functionality (for example, functionality that assists with wagering, payouts... or management of participation funds)."* Converting Game Night exclusively to a unitless **Points / Chips-Only Mode**

(*Points In, Points Out, Net Score Differential, Host Tally*) with zero currency symbols or rake fields eliminates both criminal/civil ledger exposure and Google Play rejection, enabling Riverline to declare *"no financial features"* on Google Play's Financial Features Declaration **[Judgement]**.

- **Redesign 2 — Why "Live Decision Mode" Must Be Retired and Replaced by Off-Table Study Controls:** Apple App Store Review Guideline 5.3.4 states verbatim: *"Illegal gambling aids, including card counters, are not permitted on the App Store."* Meanwhile, PokerStars' Prohibited Software Policy and GGPoker's Security & Ecology Policy (SEP § 2) prohibit keeping any solver, equity calculator, or chart lookup tool open during active play:

  > *"2.1. RTA is defined as any external assistance which provides users with an unfair advantage over their opponents, influences their decision making in real-time, or allows them to play in a manner they otherwise would not be able to replicate... 2.3. Referring to charts of any kind during play is strictly prohibited. 2.4. Potential actions for breaches: PERMANENT BAN, CONFISCATION OF FUNDS."* — *GGPoker Security & Ecology Policy § 2*

  Retiring the name *"Live Decision Mode"* in favor of **"Off-Table Scenario Simulator"** / **"Post-Session Spot Lab,"** requiring full post-hand context entry, adding a first-launch Anti-RTA click-through banner, anonymizing shared hand links (GGPoker SEP § 7.3), and pausing desktop solver/chart views when `PokerStars.exe`, `GGnet.exe`, or `888poker.exe` are detected in process memory aligns Riverline with both Apple § 5.3.4 and operator compliance standards **[Judgement]**.

- **Redesign 3 — Israeli Amendment 13 Privacy, Section 14I Subscription Cancellation, and Russian 149-FZ / 152-FZ Isolation:** In Israel, Amendment 13 to the Protection of Privacy Law, 5741-1981 (effective August 14, 2025, with immediate PPA administrative fines demonstrated in the August 2026 *A.D Karting Hutzot* enforcement action) requires a 5-element Section 11 Notice at sign-up and an internal Database Definitions Document (מסמך הגדרות מאגר), while Sections 13D(c) and 14I of the Consumer Protection Law mandate a prominent homepage **"Cancel Subscription / ביטול עסקה"** link with 3-business-day termination. In the EU, strictly necessary `localStorage` requires no cookie consent banner under ePrivacy Directive 2002/58/EC Art. 5(3), and microenterprises (<10 staff, ≤ €2M turnover) are exempt under European Accessibility Act 2019/882 Art. 4(5) (paralleling Israel's under-ILS 100,000 turnover exemption in IS 5568 Regulation 35C / accessiBe). In Russia, to avoid 24-hour extrajudicial domain blocking by Roskomnadzor under Federal Law No. 149-FZ Art. 15.1(5)(1)(d) and Law No. 244-FZ or data-localization fines under Law No. 152-FZ Art. 18(5), the Russian (`RU`) interface must display an `18+` badge (Law No. 436-FZ), contain zero links to poker operators or currency symbols, and operate strictly local-only without storing Russian citizens' PII in foreign cloud databases **[Judgement]**.

### 7.2 Copy-Paste Ready Multilingual Disclaimers & Israeli Section 11 Notice [Judgement]

To operationalize the statutory and platform-policy controls across English (`EN`), Hebrew (`HE`), and Russian (`RU`), Riverline must surface standardized, copy-paste-ready legal disclosures at three distinct user touchpoints: (1) a first-launch click-through modal and persistent footer link displaying the **18+ Educational Study-Tool, Anti-RTA & Points-Only Scorekeeper Disclaimer**; (2) an inline **Section 11 Privacy Protection Notice** presented prior to any optional account registration or cloud synchronization under Amendment 13 to the Israeli Protection of Privacy Law, 5741-1981 and GDPR Article 13; and (3) a strict local-storage declaration in the Russian interface confirming that zero personal data is transmitted to foreign servers under Russian Federal Law No. 152-FZ **[Judgement]**. The table below provides the complete trilingual wording for immediate integration into the application repository.

| Disclosure Type | Copy-Paste Ready Text (EN · HE · RU) [Judgement] |
|---|---|
| **1A. English (EN) Clause 1 & Clause 2 (18+ Non-Gambling & Anti-RTA Ban)** | **18+ Educational Study Tool Only — No Real-Money Gambling & Strict Fair-Play Policy:** (1) *No Gambling or Real-Money Play:* This application is an independent mathematical game-theory and strategy study workstation intended solely for adults aged 18+ (or 21+ where required by local law). It does not offer, facilitate, or promote real-money gambling, wagering, or prizes of real-world value. No virtual points or chips can be purchased, wagered for value, transferred, or redeemed for money or goods. (2) *Prohibition on Real-Time Assistance (RTA):* This workstation is designed strictly for off-table study and post-session hand review. Using strategy charts, equity calculators, or scenario simulators during live or online poker play is strictly prohibited by online poker operators (including PokerStars and GGPoker) and live cardroom rules, and may result in permanent account bans and confiscation of funds. You agree never to use this application while any poker client or live game is in progress. |
| **1B. English (EN) Clause 3 (Points-Only Scorekeeper & No Debt Enforcement)** | (3) *Friendly Scorekeeper Only — No Financial or Debt Enforcement:* The Game Night organiser is a unitless points and chip-count tally for private, recreational social games where lawful. The application does not process payments, hold funds, act as a banker, or verify the legality of any private game. Score summaries are for recreational scorekeeping only and do not create, evidence, or constitute a legally enforceable debt, loan, or financial obligation (including under Section 32 of the Israeli Contracts Law, 5733-1973, and Section 230 of the Israeli Penal Law, 5737-1977). |
| **2A. Hebrew (HE) 18+ Study-Tool, Anti-RTA & Points-Only Scorekeeper Disclaimer** | (1) **כלי לימוד ומחקר מתמטי בלבד (18+) — ללא הימורים בכסף אמיתי ומדיניות משחק הוגן:** *איסור הימורים:* יישום זה הינו סביבת לימוד וניתוח מתמטית עצמאית לתורת המשחקים, המיועדת לבגירים מעל גיל 18 בלבד. האפליקציה אינה מציעה, מאפשרת או מעודדת משחקים אסורים או הימורים בכסף אמיתי (כהגדרתם בחוק העונשין, התשל"ז-1977). לנקודות או לאסימונים הווירטואליים באפליקציה אין כל ערך כספי ולא ניתן לרכוש, להמר או לפדות אותם. (2) *איסור סיוע בזמן אמת (RTA):* האפליקציה מיועדת ללימוד מחוץ לשולחן המשחק ולניתוח ידיים לאחר סיום המשחק בלבד. חל איסור מוחלט להשתמש באפליקציה בזמן משחק פוקר פעיל. (3) *מחשבון ניקוד חברתי בלבד:* כלי "ערב המשחקים" משמש לניהול ניקוד יחסי ללא ערך כספי במשחקי חברה פנאיים כחוק (סעיף 230 לחוק העונשין). רישומי הניקוד אינם מהווים חוב, הלוואה או התחייבות כספית הניתנת לאכיפה (בהתאם לסעיף 32 לחוק החוזים (חלק כללי), התשל"ג-1973). |
| **2B. Hebrew (HE) PPL Amendment 13 Section 11 Privacy Notice** | **הודעת פרטיות לפי סעיף 11 לחוק הגנת הפרטיות, התשמ"א-1981 (תיקון 13):** בשימושך במצב מקומי, כל נתוני הלמידה נשמרים בדפדפן על מכשירך בלבד (`localStorage` / `IndexedDB`). במידה ותבחר לפתוח חשבון סנכרון אופציונלי: (א) מסירת המידע תלויה ברצונך החופשי ואינה חובה חוקית (ללא חשבון, האפליקציה פועלת במלואה באופן מקומי); (ב) מטרת האיסוף היא סנכרון נתוני לימוד וניהול מנוי בלבד; (ג) בעל השליטה במאגר המידע: ויקטור סטז'קו (דוא"ל ליצירת קשר יופיע בעמוד מדיניות הפרטיות); (ד) המידע לא יועבר לצדדים שלישיים למעט ספקי תשתית וסליקה מורשים; (ה) עומדות לך זכויות העיון, התיקון והמחיקה לפי סעיפים 13 ו-14 לחוק. |
| **3. Russian (RU) 18+ Study & Local-Only Notice (149-FZ / 152-FZ / 436-FZ)** | **18+ Образовательный математический тренажер — Без азартных игр и ставок:** Данное приложение является исключительно образовательным математическим калькулятором и тренажером по теории игр для лиц старше 18 лет (Федеральный закон № 436-ФЗ). Приложение не организует, не проводит и не рекламирует азартные игры на деньги или иное имущество (Федеральный закон № 244-ФЗ и ст. 15.1 Федерального закона № 149-ФЗ). Все очки и фишки в приложении являются условными математическими единицами без денежной стоимости. Использование приложения во время активной игры (RTA) строго запрещено. Все пользовательские данные в русскоязычной версии хранятся исключительно локально в браузере на устройстве пользователя (`localStorage` / `IndexedDB`) без сбора и передачи персональных данных на внешние серверы (Федеральный закон № 152-ФЗ). |

## 8. Unified Master Execution Roadmap (Now / M1 / M2 / M3) and Verified Source Registry

Combining the repository's live engineering sequence (CURRENT_PHASE.md and ROADMAP.md) with the actionable deliverables of all six research tracks yields a four-stage execution roadmap **[Judgement]**:

| Phase | Core Product & UI Tickets | Solver & Verification Pipeline | Brand, Audio & Legal Gate |
|---|---|---|---|
| **Phase 0 / NOW (Weeks 1–2)** | • Human QA acceptance on `DS-FOUNDATION-001`, `SHELL-001`, `SAVED-COMPOSITION-002`, and `ANALYZE-COMPOSITION-001` (`QA-ANALYZE-COMPOSITION-005`).<br>• Complete `PREFLOP-SANITY-001` (suppress heuristic AA/premium fold leakage) and `BLUFF-001`. | • Extend b-inary/poker-cfr (BSD-2) to 6–9 seats with card bunching to ship **`Verified · Equilibrium`** ≤ 15 bb push/fold tables.<br>• Stand up the LBR/BR + AIVAT (pluribus) + 40-invariant `litmus` (robopoker) verification harness. | • Select final public brand name from the 12 vetted Low-Risk candidates.<br>• Replace any "Live Decision Mode" copy with **"Off-Table Scenario Simulator"**.<br>• Lock Game Night spec to unitless **Points-Only Mode**. |
| **M1 (Weeks 3–4): Trustworthy Base** | • Complete `QA-BETA-DESIGN-CORRECTION-002` (desktop/theme matrix, lag profiling) and bounded Beta Repair Sweep.<br>• Finalize 31-moment motion layer and trust/provenance badges across all views. | • Inspect `src/preflop.rs` in exinori/DCFR-SOLVER via local `git clone` (resolving the **UNVERIFIED** leaf question).<br>• Benchmark 8–9 player factorized card-bunching approximation within 0.2% equity. | • Integrate 100% CC0 **Set A + Set B** real foley audio files (16-bit / 44.1 kHz, −3 dB shelf above 5.5 kHz).<br>• Attach EN/HE/RU 18+ Study-Tool & Scorekeeper Disclaimers + OSS attributions. |
| **M2 (Weeks 5–8): Core Reworks** | • Resume Personal Strategy at `RANGE-CAL-002B/C/D` (holdout confidence validation, `Teach Riverline Next` boundary queue, inferred Matrix) and `PERSONAL-STRATEGY-CONTINUATION-001C`.<br>• Deliver Post-Hand "Key Decisions" Review with 1-click **Retry**, `Advanced Equity v1`, Points-Only Game Night, and Table Rules (`Home 0-rake` + `ClubGG 0.1bb/seat`). | • Run fixed-size 6–9 seat preflop Linear MCCFR (pluribus / DCFR-SOLVER) bridged to 47-flop offline b-inary/postflop-solver leaf solves to promote 30–60 bb (0-rake) and 100 bb (raked) RFI/3B/4B/BvB to **`Verified · Equilibrium`**.<br>• Solve 30–60 bb limped pots via `--rnr-model` as **`Verified · vs. Model`**. | • Publish all 7 legal disclosures (including 5-element Israeli PPL Section 11 Notice, Footer `localStorage` notice, and `IS 5568` Accessibility Statement).<br>• Add Electron process detection (`PokerStars.exe`, `GGnet.exe`) and shared-hand auto-anonymization. |

| Phase | Core Product & UI Tickets | Solver & Verification Pipeline | Brand, Audio & Legal Gate |
|---|---|---|---|
| **M3 (Weeks 9–14): Expansion & Release** | • Launch local `reference-pack/v1` (`REFERENCE-PACK-001` with SHA-256 verification), Orbit sessions, connected workspaces, richer Home hub, calm teacher mascot Chippy, and installable PWA / responsive phone views. | • Train 6–8 handed multiway flop blueprints (24–64 EMD buckets) + 1-street flop resolver (`src/flop.rs`) for top 20–30 3-way flop nodes.<br>• Optional HRC Pro v4.0+ ($359.90/yr) 16,384-bucket reference pack export.<br>• Optional 2–4 week clean-room Turn/River DCFR subgame recalculator. | • Obtain Israeli gaming/privacy/tax counsel sign-off; register *Osek Patur / Osek Murshe*; add homepage **"Cancel Subscription / ביטול עסקה"** link (CPL § 14I); onboard Paddle/Lemon Squeezy.<br>• Pass Apple (§ 5.3.4) & Google Play store review. |

## 8.1 Complete Verified Reference & Primary Source Registry

1. **Riverline Internal Repository Authorities:** `RIVERLINE_BRIEF.md` (2026-10-21), AGENTS.md, CURRENT_PHASE.md, ROADMAP.md, and RIVERLINE_MASTER_PRODUCT_STRATEGY_PLAN.md.
2. **Riverline Research Archive (Google Drive, Oct 21–23, 2026):** Track 1: Poker Solver Licensing and Quality; Track 2: Free Preflop & Multiway Solver Architecture; Track 3A: Brand & Naming Audit & Track 3B: Supplemental Naming Audit; Track 4: Competitor & UX Survey 2026; Track 5: Audio & Foley Source Licence Audit; Track 6: Regulatory, Platform-Policy & Anti-RTA Compliance.
3. **b-inary Postflop & Push/Fold Solvers:** b-inary/postflop-solver, Cargo.toml (AGPL-3.0-or-later), Issue #46 (Suspension Notice), desktop-postflop, wasm-postflop, and b-inary/poker-cfr (BSD-2-Clause).
4. **Open-Source Multiway & Preflop Solvers:** conorarmstrong/pluribus (NoRegrets, MIT), exinori/DCFR-SOLVER (MIT), krukah/robopoker (MIT), MatthewPDingle/GTOpen (No License File), bupticybee/TexasSolver (AGPL-3.0), google-deepmind/open_spiel (Apache-2.0), ericgjackson/slumbot2019 (MIT), uoftcprg/pokerkit (MIT), andrewprock/pokerstove (BSD-3-Clause), ozzi7/Poker-MCCFRM, fedden/poker_ai, datamllab/rlcard, EricSteinberger/Deep-CFR, and deepfold.co Free Poker Solver 2026.
5. **Commercial Solver EULAs & ToS:** HoldemResources Calculator v4.0+ EULA, HRC Pricing, PioSOLVER Licensing Terms, PioSOLVER Business FAQ, GTO+ EULA, Simple Poker Terms & Conditions, MonkerWare Terms of Service, GTO Wizard Terms, GTO Wizard Benchmarking API Terms, and FSF GPL FAQ (#WhatCaseIsOutputGPL).
6. **Game-Theory & CFR Literature:** Brown & Sandholm, Solving Imperfect-Information Games via Discounted Regret Minimization (AAAI 2019); Brown & Sandholm, Superhuman AI for Multiplayer Poker (*Science* 2019); Brown et al., Deep Counterfactual Regret Minimization (ICML 2019); Lanctot et al., Monte Carlo Sampling for Regret Minimization (NeurIPS 2009); Ganzfried & Sandholm, Potential-Aware Imperfect-Recall Abstraction (AAAI 2014).
7. **Brand, Domain & Trademark Sources:** Riverline Poker, Riverline Poker House Rules, Riverline AI, Riverline Resolution, Riverline on Google Play, Class 42 Application #6935248, USPTO #98044162, USPTO #78843317, riverline.com (NJ Transit), Revyline Russia, River Noodle Bar Israel, and rebar Israel.
8. **Competitive & EdTech Benchmarks:** GTO Wizard Help, TwoPlusTwo Live Cash Thread, r/poker GTO Wizard Thread, GTO Gecko 2026 Survey, GTOBase, Postflop+, FlopzillaPro, PokerSnowie AI Weaknesses, Upswing Lab, RIO Vision FAQ, Hand2Note Manual, Live Poker Trainer, Chess.com Game Review, Chess.com CAPS2, and Brilliant.org Help.
9. **Audio & Foley Archives:** Freesound CC0/CC-BY FAQ, Kenney Casino Audio, OpenGameArt Casino SFX, BMacZero Playing Card Sounds, Sonniss GDC License, BBC RemArc Licence, ZapSplat Standard License, and Mixkit License.
10. **Statutory, App-Store & Anti-RTA Authorities:** Israeli Penal Law §§ 224–235, Chambers Gaming Law Israel 2025, C.A. 476/17 *Rafi Amit*, Israeli Contracts Law § 32(a), Israeli PPL Amendment 13 (§ 11), INPLP Amendment 13 Fines, Israeli Consumer Protection Law § 14I, IS 5568 Accessibility, Apple Guideline 5.3.4, Google Play Real-Money Gambling Policy, PokerStars Prohibited Software, GGPoker SEP § 2, and Russian Federal Law No. 149-FZ Art. 15.1.